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EVALUATING THE EQUAL-INTERVAL HYPOTHESIS WITH TEST SCORE SCALES

BEN DOMINGUE

INSTITUTE OF BEHAVIORAL SCIENCE, UNIVERSITY OF COLORADO BOULDER

The axioms of additive conjoint measurement provide a means of testing the hypothesis that testing data can be placed onto a scale with equal-interval properties. However, the axioms are difficult to verify given that item responses may be subject to measurement error. A Bayesian method exists for imposing order restrictions from additive conjoint measurement while estimating the probability of a correct response. In this study an improved version of that methodology is evaluated via simulation. The approach is then applied to data from a reading assessment intentionally designed to support an equal-interval scaling.

Key words: conjoint measurement, Rasch model, interval scale.

Discussion about whether psychological variables can be intervally scaled dates to the beginning of the 20th century (e.g. Hölder, 1901; Campbell, 1933; Ferguson, Myers, Bartlett, Banister, Bartlett, Brown, et al., 1940). This topic merits renewed focus in the context of educational testing (for variables like math and reading ability) since such scores are increasingly important in education policy in the United States as well as Europe. For example, Briggs (2013) discusses the example of tests that have been vertically scaled for the purpose of making statements about student growth in terms of magnitudes. The validity of such statements hinges upon whether the underlying scale has equal-interval properties. Additive conjoint measurement (Luce & Tukey, 1964; Krantz, Luce, Suppes, & Tversky, 1971) offers a framework that can be used to evaluate whether a manifest or latent variable has a quantitative structure that would support equal-interval scaling. Suppose that a variable depends upon two other variables in the same way density depends upon weight and volume. In education, the probability of a correct response to a test item is frequently thought, as in the Rasch (1960) formulation, to depend solely upon student ability and item difficulty. The theory of conjoint measurement proves that if certain axioms are true, then interval scales are known to exist for all three variables. The theory, however, does not provide a proscriptive method for scale construction. Many of the axioms require the variables to obey certain orderings. Since the observed item responses in educational tests are typically assumed to contain some degree of measurement error, it can be difficult to know whether the failure of an ordering to hold is real or due to chance. This challenge was initially addressed by Karabatsos (2001). However, there are some weaknesses in that approach that can be improved upon; this is the focus of the present study.

The Rasch (1960) model is frequently used to create scales with data from educational tests. Since the Rasch model can be conceptualized as a probabilistic formulation of additive conjoint measurement (Keats, 1967; Fischer, 1968; Brogden, 1977; Karabatsos, 2000), data that fits the Rasch model should allow for interval scaling. More complex item response models such as the three-parameter logistic (3PL) model (Birnbaum, 1968) do not have this property, so if interval scales are desired then the Rasch model is the most natural choice for an item response model. However, judging whether the Rasch model fits a given dataset is not straightforward since there are a number of fit statistics (see Glas & Verhelst, 1995) and choosing critical values is not always straightforward due to dependencies on sample size (Wu & Adams, 2013). The methodological

Requests for reprints should be sent to Ben Domingue, Institute of Behavioral Science, University of Colorado Boulder, Boulder, CO, USA. E-mail: ben.domingue@gmail.com

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approach presented here would allow for judgment about the possible scale type (interval versus ordinal) that may be achieved by an estimation procedure *prior* to estimation. That is, given any set of dichotomously scored responses, a judgment can be made whether the structure of the data is sufficient to allow for interval scaling before efforts are made to actually scale the data.

The paper is organized as follows. Section 1 discusses additive conjoint measurement before Sections 2 and 3 describe the original and updated methodology. Section 4 presents the results of an application of the updated methodology to simulated data. This exercise demonstrates that the methodology is sensitive to specific deviations from the axioms. This is an important result since earlier research did not explore the efficacy of the methodology. Section 5 presents results using empirical data. Earlier work (Kyngdon, 2011) had found this particular data to be consistent with the axioms, but the opposite conclusion is reached here in a more thorough analysis. Finally, Section 6 contains a discussion and suggestions for future research.

1. Additive Conjoint Measurement

Additive conjoint measurement (ACM) demonstrates that when one variable can be expressed as an additive function of two other variables, values for all three variables can be placed onto a scale with interval properties. Write this as $z_{ij} = f(x_i, y_j)$ where $z_{ij} \in Z$, $x_i \in X$, $y_j \in Y$, and $f : X \times Y \to Z = \mathbb{R}$. The function f is restricted by the fact that it must be non-interactive, meaning that f can be decomposed as $g(x) + h(y)$. Transformations of f to meet this requirement are permissible. For example, if $f(x, y) = xy$ then $\log xy = \log x + \log y$ yields an acceptable decomposition. Any combination involving xy that cannot be split means that f is interactive. Interval scales for X, Y, and Z exist when certain axioms hold with respect to the three variables.

In the context of educational testing, let X be a set of individuals and Y a set of items. Let $f(x_i, y_j) \geq f(x_m, y_n)$ indicate that the probability of a correct response for individual x_i to item y_j is larger than the probability of a correct response for individual x_m to item y_n. Consider the following axioms on the triple $(X, Y, \geq)$ where x_1, x_2, x_3 are individuals from the set X and y_1, y_2, y_3 are items from Y:

1. Independence (or Single Cancellation).

$$\text{If } f(x_1, y_1) \geq f(x_2, y_1); \quad \text{then } \forall y_2 \in Y,\ f(x_1, y_2) \geq f(x_2, y_2).$$

   Thus, there is an induced order on X and one can unambiguously write $x_1 \geq x_2$. An analogous condition must also hold for Y. The independent ordering of rows and columns can be either assumed as in Krantz et al. (1971) or established as a byproduct of solvability and double cancellation as in Michell (1990).
2. Double Cancellation.

$$\text{If } f(x_1, y_2) \geq f(x_2, y_1) \quad \text{and} \quad f(x_2, y_3) \geq f(x_3, y_2), \quad \text{then } f(x_1, y_3) \geq f(x_3, y_1).$$

   Additional discussion of this axiom is provided below.
3. Solvability.

$$\forall x_1 \in X, y_1, y_2 \in Y, \exists x_2 \in X \quad \text{such that } f(x_1, y_1) = f(x_2, y_2).$$

4. Archimedean Condition. Michell (1990) describes this condition as follows: "no value of a quantitative variable is infinitely larger than any other value" (p. 73). In essence, this condition ensures comparability between any two values as they can only be a finite distance apart.

If these axioms hold, then $(X, Y, \geq)$ is an additive conjoint structure. That these are sufficient conditions for the existence of interval scales was shown by Luce and Tukey (1964, p. 10, see Theorems VID through VIJ). It was first shown that for such additive conjoint structures there exist functions $\phi_1 : X \to \mathbb{R}$ and $\phi_2 : Y \to \mathbb{R}$ (unique up to linear transformation) such that

$$f(x_1, y_1) \geq f(x_2, y_2) \quad \Leftrightarrow \quad \phi_1(x_1) + \phi_2(y_1) \geq \phi_1(x_2) + \phi_2(y_2). \tag{1}$$

This is convenient since it allows for operation with the simpler ϕ functions, rather than f, to create dual standard sequences. In classical measurement theory, standard sequences are generalized versions of the demarcations of inches on a ruler. Dual standard sequences are the extension of this concept to the conjoint measurement case. For $m, n \in \mathbb{Z}$, (x_m, y_n) is a dual standard sequence if $f(x_i, y_j) = f(x_p, y_q)$ whenever $i + j = p + q$. A simple case will help to illustrate the point. If $i = q = 1$ and $j = p = 2$, then we have $f(x_1, y_2) = f(x_2, y_1)$. Expressed in the educational testing context, a correct answer from person 1 to item 2 is just as likely as a correct answer from person 2 to item 1. Luce and Tukey (1964) showed that if (x_i, y_j) is a dual standard sequence then

$$\phi_1(x_n) - \phi_1(x_0) = n\big[\phi_1(x_1) - \phi_1(x_0)\big]$$
$$\phi_2(y_n) - \phi_2(y_0) = n\big[\phi_2(y_1) - \phi_2(y_0)\big].$$

One can now clearly see that the measures implied by ϕ_1 and ϕ_2 are interval. A move from x_0 to x_n is equivalent to n moves from x_0 to x_1. The distance from x_0 to x_1 is acting as a standard unit against which other moves in X can be compared. Construction of the dual standard sequence may be quite challenging, but if the axioms hold then a construction, involving manipulations of elements in X and Y, is possible. In psychometrics, manipulation of X and Y is difficult when compared to a field such as psychophysics, where researchers can manipulate variables such as sound intensity and frequency. Work on item difficulty manipulation has a history dating back to the early 1970s (Scheiblechner, 1972), but a modern example is the Lexile Framework for Reading (Stenner, Burdick, Sanford, & Burdick, 2006; Stenner, Smith, & Burdick, 1983). Using a theory that the Rasch difficulty of reading comprehension items is a function of sentence length and word frequency, they are able to estimate the difficulty of a newly constructed item *prior* to its administration (taking advantage of a database of prior students' responses to Lexile items). Manipulation of individual abilities is more challenging, but perhaps possible (e.g., Briggs, 2013).

Returning to the axioms, emphasis is on verifying the first two axioms since the last two are impossible to test directly in most cases. Paraphrasing Michell (1990, p. 79), finding elements to satisfy solvability might be impractical due to time or money constraints. The steps required to show that a certain value is bounded (and thus restricted by the Archimedean condition) may be practically impossible, analogous to trying to demonstrate that the sun is a finite distance from the earth using a common 1 foot ruler. Scott (1964) developed testable indirect conditions that are necessary but not sufficient conditions for Axioms 3 and 4 to hold but these conditions are not used here.

Reconsider double cancellation. Assume that $(X, Y, \geq)$ is an additive conjoint structure and that the antecedent in the axiom of double cancellation is true. This implies that

$$\phi_1(x_1) + \phi_2(y_2) \geq \phi_1(x_2) + \phi_2(y_1),$$
$$\phi_1(x_2) + \phi_2(y_3) \geq \phi_1(x_3) + \phi_2(y_2).$$

Summing the above inequalities produces:

$$\phi_1(x_1) + \phi_2(y_2) + \phi_1(x_2) + \phi_2(y_3) \geq \phi_1(x_2) + \phi_2(y_1) + \phi_1(x_3) + \phi_2(y_2).$$

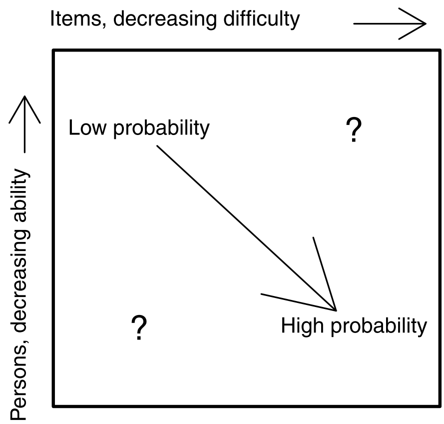

FIGURE 1.
Rendering of row and column ordering.

Canceling terms leads to

$$\phi_1(x_1) + \phi_2(y_3) \geq \phi_1(x_3) + \phi_2(y_1)$$

implying that $f(x_1, y_3) \geq f(x_3, y_1)$. Thus, double cancellation is a natural consequence of the additivity of the structure.

Consider Figure 1 which demonstrates the ordering of rows and columns to be used throughout. Columns are ordered by decreasing difficulty so that the most difficult items are at the left. Rows are ordered by increasing ability so that the lowest ability individuals are at the top. As a consequence, cells with the lowest probability of a correct response are at the top left and the highest probabilities at the bottom right. Moving on any path down and right through the cells, the probability of a correct response *must* be increasing if single cancellation holds. However, moving at right angles to this major diagonal leads to uncertain changes in the probabilities. In the top right is the probability of correct responses for low ability individuals to easy items. At bottom left, the probability for high ability individuals to hard items. The effect of double cancellation is to ensure that certain orderings, those needed to prove the existence of interval scales, hold along the minor diagonals.

1.1. Ordering Along Minor Diagonals

Consider the probability of a correct response from the Rasch model. If P_{ij} refers to a true underlying probability for individual i and item j, then it will always be the case that $P_{ij} < P_{i+1,j+1}$, assuming the ordering from Figure 1. There is no such simple statement that can be made about the minor diagonal in the context of the sub-matrix shown in Figure 1. The double cancellation axiom exists to ensure some reasonable degree of behavior, but it is illustrative to see how unordered the minor diagonal can be before moving forward.

Consider the probability of a correct response generated from the Rasch model using item difficulties 1.561, 1.236, 1.071 and person abilities −0.323, 0.115, 0.203. Here is the 3×3 matrix of probabilities formed on the basis of those parameters:

$$\begin{array}{ccc} 0.132 & 0.174 & 0.199, \\ 0.191 & 0.246 & 0.278, \\ 0.205 & 0.263 & 0.296. \end{array} \tag{2}$$

It is easy to verify the ordering along the major diagonal, but there is no definite order along the minor diagonal here. Clearly $P_{31} < P_{22}$ and $P_{22} > P_{13}$. In fact, $P_{31} > P_{13}$. The order along the minor diagonals is important since these orderings are of interest in checking double cancellation. To consider the canonical form of this axiom, begin with the inequalities:

$$P_{21} < P_{12},$$
$$P_{32} < P_{23}.$$

If these two logical predicates have the same truth value (either both true or both false), then a certain ordering must hold with respect to the $(1, 3)$ and $(3, 1)$ corner cells. Since here they have different truth values, no ordering of the corners is necessary. It is crucial that one keep in mind the fact that the minor diagonals are, in general, unordered. The relatively weak (when compared to the strict ordering of the major diagonal implied by single cancellation) requirements of double cancellation are the only necessary orderings.

2. Methodology

The methodology introduced here grows out of a number of articles suggesting methods of checking the axioms in a testing context. A small amount of terminology will be useful. A conjoint matrix denotes the matrix formed by aggregating individuals at a common ability such that a cell gives the proportion of individuals at that ability who answered a certain item correctly. A 3-matrix describes a 3×3 matrix formed by the choice of three rows and three columns from a conjoint matrix. How 3-matrices are chosen from a conjoint matrix to check for consistency with the axioms *and* the number of 3-matrices to check are both important issues, to be discussed later.

The first attempt to check the ACM axioms in the same vein as this study was Perline, Wright, and Wainer (1979). In that study, the authors checked the axioms using observed proportions of correct responses. One noteworthy aspect of this work is that they test all possible 3-matrices formed from a conjoint matrix for double cancellation. Given the computational resources available in 1979, checking this many 3-matrices was only possible since they had relatively small set of items. In two empirical examples, they found violations in 5 % and 10 % of the checks. Another approach was described in Green (1986) which was influenced by the work of Iverson, and Falmagne (1985) in that the author attempts to allow for measurement error in the observed proportions. The method is relatively straightforward. Frequentist confidence intervals for the probability of observing a correct response in each cell of a conjoint matrix are computed and then used to test whether differences between cells hold at some specified significance level. Using this technique, Green determined that single and double cancellation were both violated using data from 19 multiple-choice (general knowledge) items given to over 900 undergraduates. Further work on the frequentist approach can be found in Davis-Stober (2009).

This paper uses Bayesian, instead of frequentist, techniques that originate in Karabatsos (2001). The key idea behind the Bayesian approach is to enforce the axioms stochastically via a Metropolis–Hastings jumping distribution and then to determine whether the resulting estimates for the probability of a correct response are reasonable given the observed data. Karabatsos reanalyzed the data originally used by Perline et al. (1979) but now allowing for measurement error. His analyses were meant to be illustrative and did not modify the original findings substantively, but the methodology he used deviated substantially from the techniques of Perline et al. (1979).

Suppose the responses to a set of items by a set of persons are functions of a unidimensional latent variable. Let P be the $I \times J$ matrix that contains the true response probabilities. Each cell in the conjoint matrix, denoted $\hat{P}^{\mathrm{MLE}}$ with dimensions $I \times J$, contains the percentage of respondents with a certain ability who answered the appropriate item correctly. $\hat{P}^{\mathrm{MLE}}$ is only an estimate of P due to measurement error. This estimator is in fact the maximum likelihood estimator for P since we are not imposing any response model beyond the assumption that individual responses are Bernoulli random variables.

To determine whether the axioms are true for P, the order restrictions inherent in the cancellations axioms are imposed stochastically via the jumping distribution. This idea was first suggested by Devroye (1986) and adapted to the MCMC context in Gelfand, Smith, and Lee (1992). Here a Metropolis–Hastings (MH: Metropolis, Rosenbluth, Rosenbluth, Teller, & Teller, 1953; Hastings, 1970) algorithm is used. The MH algorithm estimates parameters by taking a random walk through the parameter space. This random walk has the property that the values at time $t+1$ depend upon only the values at time t (hence it is a Markov chain). Given the values in the chain at time t, a draw from the "jumping distribution" (also known as the proposal) yields the proposed value for the next step. The jumping distribution is typically a multivariate normal distribution centered at the values from time t with a suitably defined covariance matrix. The proposal value is accepted or rejected with probability based on the ratio of the posterior distribution at the proposal and current (time $t-1$) values. Due to the possibility of rejection, the chain contains "steps" that are stationary. Additional details on the MH algorithm can be found in many texts (e.g., Jackman, 2009; Gelman, Carlin, Stern, & Rubin, 2004).

Before describing the algorithm that considers both single and double cancellation, consider a simpler example involving only single cancellation. Concentrate on $P_{(i,j)}$ with $1 \le i \le I$ and $1 \le j \le J$. Under the same assumptions on the "direction" of row ordering that was shown in Figure 1, single cancellation implies that $P_{(i-1,j)} < P_{(i,j)} < P_{(i+1,j)}$ for each j. An initial maximum likelihood estimate for P_{ij} is readily computed from the observed data using the proportion of students at a given ability level i that answered item j correctly. The MH algorithm is now used to find an alternative estimate for P_{ij}. Consider only column j. If one fixes $\hat{P}^t_{(0,j)} = 0$ and $\hat{P}^t_{(I+1,j)} = 1\ \forall t$, then the jumping distribution for cell (i, j) is

$$\mathrm{Unif}\big[\hat{P}^t_{(i-1,j)}, \hat{P}^{t-1}_{(i+1,j)}\big] \tag{3}$$

at iteration t. The dependence of $\hat{P}^t_{(i-1,j)}$ and $\hat{P}^{t-1}_{(i+1,j)}$ on t indicates that all cells are being updated simultaneously. What is of fundamental importance is that the ordering implied by single cancellation is being imposed via the jumping distribution since the choice of proposal implies that $\hat{P}^t_{(i,j)}$ will be in the interval $[\hat{P}^t_{(i-1,j)}, \hat{P}^{t-1}_{(i+1,j)}]$. A sample drawn from the proposal density is then accepted or rejected via the MH acceptance ratio (discussed in the appendix). After burn-in and thinning, a 95 % credible region is formed from the remaining $\hat{P}^t_{(i,j)}$. If each $\hat{P}^{\mathrm{MLE}}_{(i,j)}$ falls within the credible region for the appropriate cell, then the single cancellation axiom holds stochastically. If not, a violation is detected. For example, in Table 3 of Karabatsos (2001), cell (3, 1) has a posterior with 95 % credible region spanning from 0.57 to 0.725. In fact, 60 out of 82 respondents (with the same total score on the assessment) answered this item correctly so the MLE estimate would be 0.732. Since $0.732 \notin [0.57, 0.725]$, this is a slight violation.

Checking double cancellation is more intricate. Going back to Figure 1, the smallest probabilities of a correct response are found at upper left and the largest at bottom right. However, moving away from this main diagonal at right angles, it is unclear exactly how the probabilities are changing as in one direction the items are getting harder but the individuals more able and in the other the items are getting easier but the individuals less able. Karabatsos (2001, p. 414) notes that extending this method to the higher order cancellation axioms is challenging since these checks depend on the selection of a sub-matrix from the full conjoint matrix (a 3-matrix in the case of double cancellation). Looking at the choices one must make for values of i, j, and k, an additional complication seems to be that each 3-matrix presents numerous versions of double cancellation.

Writing double cancellation as

$$f(x_i, y_j) \geq f(x_j, y_i) \quad \text{and} \quad f(x_j, y_k) \geq f(x_k, y_j) \quad \rightarrow \quad f(x_i, y_k) \geq f(x_k, y_i),$$

then there are many possible instances of double cancellation that one can obtain by permuting the indices. Michell (1988) distinguishes between the strict form of double cancellation shown here, which he calls Luce–Tukey double cancellation, and a more generic form which has 36 possible instantiations from a 3-matrix. Of the 36, however, many of the forms are trivial given that single cancellation holds, so one can concentrate on a smaller subset. In fact, Michell demonstrates that it is sufficient to examine the instances:

$$f(x_1, y_2) \geq f(x_2, y_1) \quad \text{and} \quad f(x_2, y_3) \geq f(x_3, y_2) \quad \rightarrow \quad f(x_1, y_3) \geq f(x_3, y_1)$$

where $i = 1$, $j = 2$, and $k = 3$ and

$$f(x_3, y_2) \geq f(x_2, y_3) \quad \text{and} \quad f(x_2, y_1) \geq f(x_1, y_2) \quad \rightarrow \quad f(x_3, y_1) \geq f(x_1, y_3)$$

where $k = 1$, $j = 2$, and $i = 3$. Based on these two forms, the search for violations of double cancellation involves looking for data patterns 3 and 25 from Table 1 of Michell (1988). These are simply the instances in which the antecedent inequalities from the above two statements of double cancellation are true and yet the consequent is false.

Suppose a 3-matrix has already been sampled and double cancellation is to be checked. Karabatsos (2001) worked with the following jumping distribution:

$$\text{Unif}\left[\max\left\{\hat{P}^t_{(i-1,j)}, \hat{P}^t_{(i,j-1)}, \hat{P}^{t-1}_{(i+1,j-1)}\right\}, \min\left\{\hat{P}^{t-1}_{(i+1,j)}, \hat{P}^{t-1}_{(i,j+1)}, \hat{P}^t_{(i-1,j+1)}\right\}\right]. \tag{4}$$

It is easier to see via a picture:

$$\begin{array}{ccc} \cdot & \downarrow\hat{P}^t_{(i-1,j)} & \uparrow\hat{P}^t_{(i-1,j+1)} \\ \downarrow\hat{P}^t_{(i,j-1)} & \Theta & \uparrow\hat{P}^{t-1}_{(i,j+1)} \\ \downarrow\hat{P}^{t-1}_{(i+1,j-1)} & \uparrow\hat{P}^{t-1}_{(i+1,j)} & \cdot \end{array}.$$

Here Θ represents the cell whose estimate is being updated. An up arrow represents a cell whose value is greater than Θ since it is in the minimum being taking to determine the right-hand edge of the uniform jumping distribution. Similarly, the down arrow represents a cell whose value is less than Θ. Note that in this formulation, one is now ensuring that single cancellation holds for both rows and columns containing Θ. The fact that the upper-right cell has index t while the lower left cell has index $t - 1$ indicates how the chains in each cell are updated. After first updating the chain for the upper-left cell, the chain for the next cell in that row is updated. After the right-hand cell's chain for the top row has been updated, the process continues in the left-most chain of the next row. This process continues until the bottom-right cell's chain has been updated at which point it begins anew with the upper-left cell.

Kyngdon (2011) built upon Karabatsos's (2001) work by applying the methodology to a dataset more similar to those seen in operational large-scale educational tests. He also suggests a philosophical stance that is more likely to be embraced by practitioners. Karabatsos (2001) frames his work as the estimation of non-parametric item response models (p. 397) while Kyngdon views IRT models as possible *analogies* of conjoint measurement. This is motivated by his distinction between the variables one manipulates in conjoint measurement, which are (hypothesized) levels of an empirical attribute, and the estimates from IRT, which are simple numerical parameters (Kyngdon, 2011, p. 482). The non-parametric models suggested in Karabatsos's work are unlikely to be used by large-scale test developers given their extensive use at present of parametric IRT models. Given this reality, it would be useful to consider the salient axioms of conjoint measurement without requiring they leave their existing framework entirely. In considering the axioms alongside other IRT concerns, test developers can generate additional evidence about the quality of their scale.

Kyngdon's (2011) study applied the axioms of conjoint measurement to data from the Lexile Framework for Reading (Stenner et al., 2006). As already noted, this assessment system allows for the estimation of item difficulties prior to administration. Kyngdon tentatively stated that "the difficulty of reading items, as conceived of in the Lexile theory, and the reading ability of persons, are quantitative" (Kyngdon, 2011, p. 488). This was based on checking single cancellation and the two instances of double cancellation from a single 3×4 conjoint matrix (taken from the much larger 39×54 conjoint matrix of all items and abilities). Since there are many abilities and items not represented in such a small conjoint matrix that go into constructing the scale, this would at first glance seem to be an insufficient test of the equal-interval hypothesis. On the basis of previous research (McClelland, 1977), Kyngdon argued that a randomly chosen matrix was extremely unlikely to support the axioms by chance if the axioms were not supported in the conjoint matrix as a whole. He was able to verify that the cancellation axioms held in his single 3×4 matrix, but only after reversing the ordering of two items. That is, the difficulties of the items established by the Lexile theory were not consistent with single cancellation but the axioms did seem supported when he reversed the ordering. The distance between the Lexile difficulties of the two items was consistent with the error of measurement for Lexile difficulties, so this was not an especially egregious reversal. However, the current study demonstrates that the finding of an interval scale was premature.

2.1. Problems with Karabatsos's Ordering Restrictions

When checking for double cancellation within a 3-matrix, Karabatsos (2001, p. 414) proposes

$$\mathrm{Unif}\big[\max\big\{\theta^t_{(i-1)j}, \theta^t_{i(j-1)}, \theta^{t-1}_{(i+1)(j-1)}\big\}, \min\big\{\theta^{t-1}_{(i+1)j}, \theta^{t-1}_{i(j+1)}, \theta^t_{(i-1)(j+1)}\big\}\big]$$

as a jumping distribution for cell (i, j). For simplicity of notation, suppose $i = 2$ and $j = 2$. Using this notation, Karabatsos's jumping distribution is going to have the effect of ensuring that $P_{31} < P_{22}$ (due to the last value in the max) and that $P_{22} < P_{13}$ (due to the last value in the min). Consider again the set of probabilities from the Rasch model introduced earlier:

$$\begin{matrix} 0.132 & 0.174 & 0.199, \\ 0.191 & 0.246 & 0.278, \\ 0.205 & 0.263 & 0.296. \end{matrix} \tag{5}$$

If those are the present values in the set of Markov chains and interest is in forming the jumping distribution for the (3, 1) cell (current value of 0.205), the jumping distribution would be

$$\mathrm{Unif}\big[\max\{0.191, 0, 0\}, \min\{1, 0.263, 0.246\}\big] = \mathrm{Unif}[0.191, 0.246].$$

Clearly, the proposed value will be less than or equal to the value in the middle cell, 0.246. This is done to ensure that the consequent of double cancellation, $P_{31} < P_{13}$, for this 3-matrix holds. (As an aside, note that this example is not even sufficiently well ordered to use Karabatsos's approach since the value in the $(1, 3)$ cell is less than the value in the $(2, 2)$ cell.)

There are three problems here. First, the statement of double cancellation is a conditional. Recall that $P_{31} < P_{13}$ only needs to be true if $P_{21} < P_{12}$ and $P_{32} < P_{23}$. Second, one is ensuring that $P_{31} < P_{13}$ through an additional restriction on P_{22}. Forcing $P_{31} < P_{22} < P_{13}$ is not only inessential, but also not even necessarily true (as in this example). Adding these non-necessary requirements makes detection of a violation more likely but does nothing in terms of ensuring interval properties since the inequality is not required by ACM. Finally, Karabatsos only considered one of the two required forms of double cancellation. This method would never allow the $(1, 3)$ cell to be less than the $(3, 1)$ cell, a condition that is consistent with some instances of double cancellation.

Even with simulated Rasch data, it is basically impossible to know if the antecedent of double cancellation holds within any given 3-matrix and, consequently, whether the consequent needs to hold. In the simulated data, knowledge of the underlying probabilities is tied to knowledge of the true abilities. However, when response data simulated based on those true probabilities are then aggregated, the rows are now groupings via sum score. Since sum scores are not perfectly mapped to the true underlying abilities, the underlying true probability for a given cell is unknown (perhaps it would be more correct to say that there is none); and, thus, it is challenging to evaluate the truth value of statements about the ordering of cells. This is a dilemma that is resolved here using a probabilistic check of the truth of the antecedent of double cancellation before imposing that the consequent be true via the jumping distribution.

3. The Modified Approach

This section introduces several changes to the methodology from Karabatsos (2001). The goal is to establish a jumping distribution for cell (i, j) that ensures both the single and double cancellation axioms are true with respect to the estimates in the chain without imposing restrictions above and beyond those of ACM. This is a departure from Karabatsos's approach, which emphasized first the checking of single cancellation before focusing on double cancellation. There are several merits to Karabatsos's separated approach. Most importantly, it can help to specify whether axiom violations are due to violations of single or double cancellation. The motivation here behind integrating the checks of both axioms into a single step is to simplify the procedure for practitioners. Due to the complexity of the approach introduced here, a conceptual overview is given first and the precise details are presented in the Appendix. Note that the methodology discussed in the next section is predicated on a 3-matrix having already been selected from the conjoint matrix. Two approaches to selecting 3-matrices are outlined at the end of this section.

3.1. *Conceptual Overview*

Suppose a 3-matrix is to be examined. The jumping distributions will vary by location in the matrix. The middle cell is rather easy as it only has to satisfy single cancellation:

$$\begin{array}{ccc} \cdot & \downarrow & \cdot \\ \downarrow & \Theta & \uparrow \\ \cdot & \uparrow & \cdot \end{array}.$$

Recall that an up arrow represents a cell who would be larger than Θ based on how the cell is incorporated into the jumping distribution, and a down arrow represents a cell that would be smaller than Θ. In fact, the jumping distribution is fairly straightforward for all cells except $(1, 3)$ and $(3, 1)$ (the "corner" cells) since the non-corner cells only need to satisfy single cancellation. For example, cell $(2, 1)$ would have a jumping distribution based on

$$\begin{array}{ccc} \downarrow & \cdot & \cdot \\ \Theta & \uparrow & \cdot \\ \uparrow & \cdot & \cdot \end{array}.$$

TABLE 1.
Values that form a jumping distribution for a 3-matrix. Let A stand for the logical predicate $\hat{P}_{21} < \hat{P}_{12}$ and B the logical predicate $\hat{P}_{32} < \hat{P}_{23}$.

Proposal for	l_1	l_2	l_3	r_1	r_2	r_3
$\hat{P}_{11}$	–	–	–	$\hat{P}_{12}$	$\hat{P}_{21}$	–
$\hat{P}_{12}$	$\hat{P}_{11}$	–	–	$\hat{P}_{13}$	$\hat{P}_{22}$	–
$\hat{P}_{13}$	$\hat{P}_{12}$	–	$\hat{P}_{31}$[a]	–	$\hat{P}_{23}$	$\hat{P}_{31}$[b]
$\hat{P}_{21}$	–	$\hat{P}_{11}$	–	$\hat{P}_{22}$	$\hat{P}_{31}$	–
$\hat{P}_{22}$	$\hat{P}_{21}$	$\hat{P}_{12}$	–	$\hat{P}_{23}$	$\hat{P}_{32}$	–
$\hat{P}_{23}$	$\hat{P}_{22}$	$\hat{P}_{13}$	–	–	$\hat{P}_{33}$	–
$\hat{P}_{31}$	–	$\hat{P}_{21}$	$\hat{P}_{13}$[b]	$\hat{P}_{32}$	–	$\hat{P}_{13}$[a]
$\hat{P}_{32}$	$\hat{P}_{31}$	$\hat{P}_{22}$	–	$\hat{P}_{33}$	–	–
$\hat{P}_{33}$	$\hat{P}_{32}$	$\hat{P}_{23}$	–	–	–	–

Jumping distribution is Unif$[f(l_1, l_2, l_3), g(r_1, r_2, r_3)]$. Details on f and g are in the Appendix.
[a]If $A\&B$ is true.
[b]If $!A\&!B$ is true.

Consider the upper-right corner cell $(1, 3)$. There are three possible restrictions depending on the current values in the chains. Let $\hat{\theta}_{(i,j)}$ be the current values in the chain. Let A and B be the Boolean variables representing the truth values of $\hat{\theta}_{(2,1)} < \hat{\theta}_{(1,2)}$ and $\hat{\theta}_{(3,2)} < \hat{\theta}_{(2,3)}$. If A and B do not have the same truth value, then no version of double cancellation needs to hold within this cell and the restrictions on θ are simply

$$\begin{matrix} \cdot & \downarrow & \theta \\ \cdot & \cdot & \uparrow \\ \cdot & \cdot & \cdot \end{matrix}.$$

If A and B are both true, then θ needs to be greater than the value in the lower-left cell. The restrictions are then

$$\begin{matrix} \cdot & \downarrow & \theta \\ \cdot & \cdot & \uparrow \\ \downarrow & \cdot & \cdot \end{matrix}.$$

If A and B are both false, then θ needs to be less than the value in the lower-left cell. This restriction is

$$\begin{matrix} \cdot & \downarrow & \theta \\ \cdot & \cdot & \uparrow \\ \uparrow & \cdot & \cdot \end{matrix}.$$

A similar set of restrictions is placed on the lower-left-hand cell. The cells within a 3-matrix that go into forming the posterior for each jumping density within the 3-matrix are shown in Table 1. However, there are additional safeguards that must be put into place to ensure that all jumping densities are well-formed. The complete algorithm, including these safeguards, can be found in the appendix.

3.2. *Sampling of 3-Matrices*

Within a conjoint matrix, the cancellation axioms need to hold within every 3-matrix. In prior research using versions of the Bayesian approach discussed here (Karabatsos, 2001; Kyngdon, 2011), the axioms were checked within only a small number of 3-matrices. Perline et al. (1979) on the other hand checked every possible 3-matrix, but this was possible due to their consideration of relatively small conjoint matrices. For n items and m ability groupings, there are

$$\binom{n}{3}\binom{m}{3}$$

3-matrices. A test with 45 items and 40 ability groupings would have over 10^8 3-matrices. This is too large a number of 3-matrices to reasonably check all of them. In this study two approaches are contrasted. In the first approach, 3-matrices are sampled at random from the full conjoint matrix. This approach is relatively time intensive. The second approach is to consider only 3-matrices formed by adjacent rows and columns (when they have been ordered in the standard way). This approach is motivated by the observation that, in general, the most stringent single cancellation tests will be precisely those formed by considering adjacent cells.

4. Simulation Results

To ensure that the checks have statistical properties that make them reasonable tools for detecting axiom violations, this section focuses on the performance of the checks in a simulation study. Responses were simulated based on the 3PL model (Birnbaum, 1968):

$$P_{ij} = \Pr(Y_{ij} = 1) = c_j + (1 - c_j)\frac{\exp[a_j(\theta_i - b_j)]}{1 + \exp[a_j(\theta_i - b_j)]}. \tag{6}$$

The θ_i and b_j parameters are fixed across all four iterations of the simulation now described. 15,000 θ_i were sampled from the standard normal and the b_j are an evenly spaced sequence between -1.5 and 1.5. In the first iteration, $a_j = 1\,\forall j$ and $c_j = 0\,\forall j$. In subsequent iterations, an increasing number of items are no longer constrained to be Rasch items as a_j and c_j are allowed to vary. Deviant guessing parameters (the c_j) are drawn from a uniform distribution on [0, 0.3] and deviant discrimination parameters (the a_j) from a uniform distribution on [0.5, 1.5]. In iterations two through four, 10, 20, and 40 of these item-side deviations are, respectively, introduced. Since the data generating model is no longer the Rasch model, the axioms will not in general hold. Interest is in whether the checks will detect the violations of the measurement axioms caused by the deviations from the Rasch model.

The overall mean detection rates are shown in Table 2. The checks based on 3-matrices formed by adjacent rows and columns are much more stringent than those based on randomly chosen rows and columns. In the case of purely Rasch data, 2.0 % of the checks detected violations when the 3-matrices were randomly chosen compared to 18.6 % for adjacent 3-matrices. Moving to the non-Rasch data, the 3-matrices formed by randomly choosing rows and columns seemed to be more discriminating indicators of deviant items as indicated by the fact that the changes in the mean were much greater relative to the baseline detection rate than those from the adjacent 3-matrices. Compared to the baseline rate, the mean detection rate went from 2.0 % to 7.2 %, a 264 % increase. In contrast, the detection rates based on 3-matrices formed from adjacent rows and columns went from 18.6 % to 31.1 %, only a 67 % increase.

Figure 2 shows the proportion of violations for each item (the lines) as a function of ability in each iteration of the simulation (indexed by the number of non-Rasch items on the top of each panel). Since the mean detection rates may be overly sensitive to volatility at the extreme abilities, weighted means were also computed where the weights are the number of individuals at each sum score level. This approach led to even greater differentiation between Rasch and non-Rasch data for both the randomly formed and adjacent 3-matrices, but the former approach saw larger changes. The simulation demonstrates that the method is sensitive to item-side deviations from the Rasch model.

TABLE 2.
Mean detection rates.

Non-Rasch items	Random			
	Mean	% change[a]	Wt mean	% change[a]
0	2.0		1.0	
10	5.3	169	3.0	206
20	6.9	248	4.4	333
40	7.2	264	5.0	391
	Adjacent			
0	18.6		15.4	
10	28.9	56	25.5	66
20	32.9	77	29.0	88
40	31.1	67	27.5	79

[a]Relative to case where data is actually generated by Rasch model (top row).

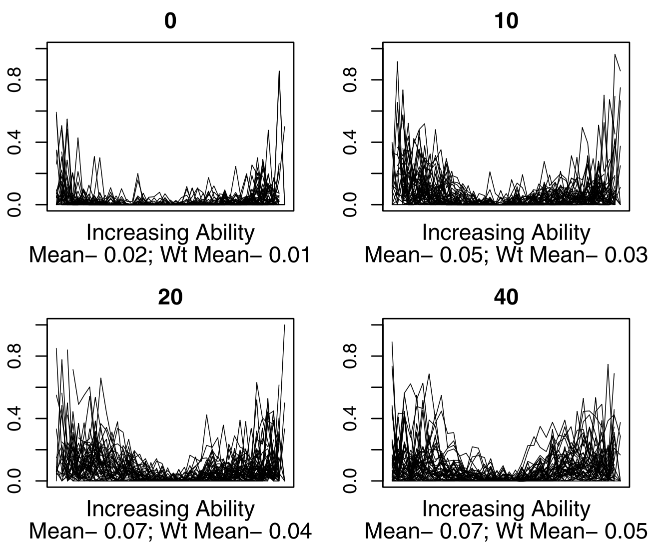

FIGURE 2.
Proportion of violations detected in each simulation using random sampling of 3-matrices.

5. Lexile Framework for Reading

MetaMetrics has incorporated an auspicious, from the perspective of ACM, design feature into their Lexile scale for reading comprehension that allows for the ordering of items in terms of difficulty prior to exposing the items to a given group of students. At the heart of this scale is a theory of reading difficulty that takes into account sentence length and word frequency (Stenner et al., 2006). Through this theory (and previous data collection), they are able to predict Rasch difficulty *before* a group of students is exposed to an item. Although the ability to manipulate both entities—in testing this would be ability and difficulty—simultaneously would be preferable, with the Lexile scale at least difficulty can be manipulated in the sense that items can be ordered by these predicted difficulties before they are administered.

After examining data generated by the Lexile assessment, Kyngdon (2011) claims there are indications (but stops short of absolute confirmation) that the data supports a scale that has interval properties:

> On the basis of these results it can be concluded that the difficulty of reading items, as conceived of in the Lexile theory, and the reading ability of persons are quantitative. A comprehensive test of the Lexile theory, however, was not the goal of the current example and therefore any judgment concerning the descriptive adequacy of this theory is premature. (Kyngdon, 2011, p. 488)

In other settings the Lexile scale has been compared with the measurement of temperature. For example: "Measurements for persons and text are now reportable in Lexiles, which are similar to the degree calibrations on a thermometer" (Stenner, 1996, n.p.).

5.1. Background on Lexile Framework

The Lexile scale is a scale of reading difficulty. Students are placed onto the Lexile scale via their performance on the "imbedded sentence cloze" items that MetaMetrics has developed. These items are described in Kyngdon (2011) as follows:

> The 'stem' of this type of reading test item is a piece of professionally edited continuous prose text taken from a published monograph. An item writer then composes a sentence that requires the respondent to select the missing word in order to 'cloze' the sentence. (p. 484)

The scale operates according to a theory of reading comprehension (Stenner et al., 2006). This theory posits that reading comprehension is a function of two quantitative features of the difficulty of the text. The first such feature is the logarithm of mean sentence length which is computed using the ratio of words in an item to the number of punctuation marks. The second feature is more complex. Words in a passage are given a frequency index, a number formed on the basis of counting the occurrences of the word in a standardized corpus (such as Carroll, Davies, & Richman, 1971). Both log mean sentence length and mean log word frequency are regressed against empirical Rasch model difficulty estimates obtained from examinees' responses to reading tests constructed of imbedded sentence cloze items (Kyngdon, 2011). When new items are developed, their difficulties can be estimated a priori by using the observed characteristics of the text and these regression coefficients. In over 15 years of work with this system, this relationship between text difficulty and the Rasch difficulty coefficients has been shown to be quite sound (Stenner, Stone, & Burdick, 2011). The difficulty of a piece of text can be evaluated using the Lexile analyzer at http://www.lexile.com/analyzer/.

The Lexile scale is a unique psychometric scale since it has a base unit: "the unit of the Lexile scale, represented by an 'L', is defined as one-thousandth of the difference in difficulty between a sample of basal primer texts and Grolier's (1986) *Encyclopedia*" (Kyngdon, 2011, p. 485, emphasis in original). On the one hand, the design of the assessment is such that items can be ordered in difficulty before administration. This would allow for the possibility of creating data specifically designed to test the conjoint axioms. On the other hand, there is nothing in either the theory or the response format to ensure that guessing does not occur or that all items have equivalent discriminations. Just because sentence length and word frequency predict difficulty, this does nothing to ensure that the items otherwise conform to the Rasch model.

5.2. Data

Data used for this study was collected in Duval, NC and first used in Kyngdon (2011). There are 9638 fourth grade students with complete response strings and 54 items. As in Kyngdon, only those sum scores between 13 and 52 (inclusive) were considered (to ensure that each sum score was represented by at least 45 students). Kyngdon used a slightly larger group of respondents, 9708 students. The specific case discussed in Kyngdon (2011) was re-analyzed using the updated methodology yielding similar results.

TABLE 3.
Lexile checks.

	Random		Adjacent	
	Mean	Wt mean	Mean	Wt mean
Original ordering	30	30	53	54
Sum scores	11	9	38	36
Pruned	9	8	30	28
Rasch baseline	2	1	18	16
State			43	41

5.3. Analysis

It is important to note that in Kyngdon (2011) the items are ordered not by sum scores but by the Lexile difficulties established via the theorized difficulty of the reading passage. This is an important point. The claim being made by Kyngdon it that the scale produced using these pre-estimated item difficulties (as opposed to item difficulties estimated from the data) has equal-interval properties. To evaluate this claim, see the top row of Table 3. In randomly formed 3-matrices, both the mean and weighted mean percentage of checks that detected violations was 30 %. For the adjacent 3-matrices, the mean and weighted mean were 53 % and 54 %, respectively. Compared to their respective Rasch baselines (from Table 2, also shown in Table 3), these are far too high. The conclusion here is obvious: the data do not seem consistent with the axioms of additive conjoint measurement. A great deal of skepticism that this data could be used to create an interval scale seems warranted.

If one instead orders items by their observed difficulty (based on sum scores) instead of the pre-determined Lexile difficulty, then there is a substantial drop in the percentage of detected violations for both the randomly formed and adjacent 3-matrices. Even these percentages are still far too large relative to what was observed for truly Rasch data for the axioms of conjoint measurement to hold. One possible explanation is that there is some ambiguity in the definition of the item difficulty since it can depend upon the specific item chosen from a range of possible items using a given passage. This is considered in Stenner et al. (2006). This study focuses on another potential explanation. The theory underlying the Lexile system does nothing to restrict guessing and discrimination to the same value for all items. While it can be argued that guessing and even discrimination (Humphry, 2010) are not strictly item-side parameters, it is informative to consider the estimates that result from fitting the 3PL model to the Duval data.

Fitting the 3PL model to the Duval data, discrimination parameters range from 0.3 to 3.4. Twenty-six items have guessing parameters above 0.2, the guessing parameter that we might expect if there are three or four distractors for each cloze item. As a comparison, the 3PL model was also fit to the responses of 48,068 students (those with missing data were dropped) to 56 items from a state assessment. This assessment is scaled with the 3PL model, which can be interpreted as an admission that the axioms of ACM should not hold. Density estimates for the Lexile and state guessing and discrimination parameters are shown in Figure 3. They are quite similar. The strict assumptions of the Rasch model are clearly too optimistic in the case of the Lexile data. The last row in Table 3 contains the percentage of violations for adjacent 3-matrices for the state data. Note that these percentages are not much larger than those from the Lexile data.

If interval scales are desired, removing certain deviant items from the Lexile data may be a possible approach. To do this, the Rasch model was fit to the data and then any item with an outfit statistic not between 0.8 and 1.2 was removed. This leaves only 33 of the original 45 items. These are probably relatively lax thresholds for outfit given the sample size (Wu & Adams, 2013), but

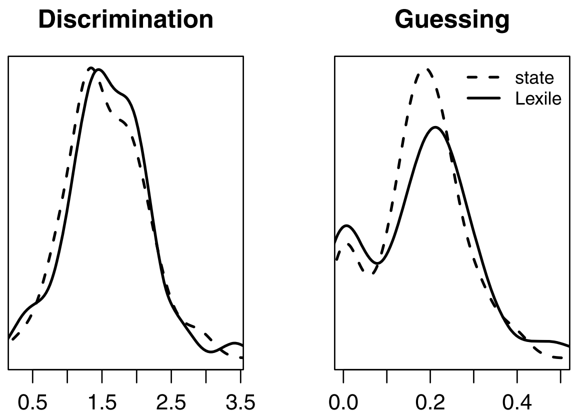

FIGURE 3.
Nonparametric density estimates for discrimination and guessing parameters in Lexile and state data.

they still illustrate the idea. After pruning, things have improved in that only 9 % of the checks (for randomly formed 3-matrices) are now showing violations. This is still far greater than what was observed in the case of simulated Rasch data. Hence, the pruned data do not seem to have sufficient structure to form an interval scale. What the checks cannot determine is the damage that could be done if one falsely assumes that a scale is interval.

6. Discussion

This paper considered a revision of the methodology first suggested in Karabatsos (2001). In particular, the jumping distribution for the Metropolis–Hastings algorithm suggested by Karabatsos was revised to check the axiom of double cancellation more accurately. The updated methodology was then used to analyze both simulated and empirical data. The use of simulated data is important since using this method without some sense for its performance in controlled settings could lead to incorrect inferences. Based on the simulation evidence, the approach is able to discriminate between data generated via the Rasch model and the 3PL model. This is to be expected given that the latter item response model, by definition, does not follow the axioms of additive conjoint measurement. Data from the Lexile Framework for Reading was also analyzed. Kyngdon (2011) concluded that the axioms held in this data. The analysis in this study led to the opposite conclusion. This demonstrated, amongst other things, the efficacy of sampling large numbers of 3-matrices when checking the ACM axioms. Since the Lexile framework is based on a theory of reading comprehension with no attention to model fit, this could be one reason for the fact that the axioms do not hold.

Code to perform the checks is freely available. Since each choice of 3-matrix requires a set of Metropolis–Hastings runs, choosing thousands of 3-matrices can be quite computationally intensive. However, this type of algorithm is “embarrassingly parallel” since it can done over multiple processors with little additional work (each 3-matrix can be handled separately in an independent process). An open source `R` (R Development Core Team, 2010) package (Domingue, 2012a) makes the work of performing analyses with dichotomously coded test data relatively straightforward.

Recent research by Luce and Steingrimsson (2011) has demonstrated that the double cancellation axiom can be replaced with different conditions. Their work was motivated by the observation from Gigerenzer and Strube (1983) that there are certain redundancies in tests of double cancellation. Luce and Steingrimsson suggest replacing double cancellation with the conjoint commutativity condition. This replacement eliminates these redundancies but would be challenging to implement in psychometrics. Roughly speaking, the basic idea is to verify that specifically chosen sets of item/individual combinations lead to equivalent probabilities of a correct response. Working with equivalencies ("indifferences" in psychophysics) would be quite challenging but may be possible in tailored applications using techniques from the item difficulty modeling literature (e.g., Scheiblechner, 1972; Fischer, 1995; Gorin, 2006).

There are a variety of potential avenues for future research. Using the method proposed here, a broader variety of data generating models could be considered alongside the 3PL. One useful strand of research would be to consider the framework of Torres Irribarra and Diakow (2012). This framework is a hierarchy of latent variable models that are based on assuming increasing structure of the latent variable (with the Rasch model being the most complex). A study that demonstrated how the violations detected by the checks varied over this hierarchy would be informative. Modifications to the methodology itself could also be explored. One natural extension would be to consider triple cancellation, a necessary condition for solvability and the Archimedean condition. Such a project would be demanding due to the quantity of 4×4 matrices that one must consider (attention would also need to be paid to the coherency conditions from Kyngdon & Richards, 2007), but it should be possible. An alternative approach would be to consider polynomial conjoint measurement. This would be an important advance since it could be used to justify interval scale interpretations when items have varying discriminations. Research on this subject exists (Kyngdon, 2011), but there are still challenges incorporating discrimination parameters into conjoint measurement. A final area of inquiry would be to study the damage that scale distortions caused by axiom violations may do to inferences based on score scales. Some research exists on this issue (e.g., Domingue, 2012b) but more work is needed.

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Appendix: Technical Notes on Methodology

The goal is to establish a jumping distribution that is consistent with the ordering restrictions of ACM. The jumping distribution will depend upon the current set of values in the set of Markov chains. Denote these current values by $\hat{\theta}_{(i,j)}$. For a given 3-matrix, each jumping distribution will have the form

$$\mathrm{Unif}\big[f(l_1, l_2, l_3), g(r_1, r_2, r_3)\big].$$

Let us first examine the $l_k, k \in \{1, 2, 3\}$. l_1 and l_2 correspond to the single cancellation checks. Set $l_1 = \hat{\theta}_{(i,j-1)}$ unless $j = 1$ in which case set $l_1 = 0$. Set $l_2 = \hat{\theta}_{(i-1,j)}$ unless $i = 1$ in which case set $l_2 = 0$. Finally, begin with a provisional value for l_3 of 0. This provisional value will possibly be altered via the rule described below. Turning to the set of values that form the right-hand end of the uniform distribution, set $r_1 = \hat{\theta}_{(i,j+1)}$ unless $j = 3$ in which case set $r_1 = 1$. Set $r_2 = \hat{\theta}_{(i+1,j)}$ unless $i = 3$ in which case set $r_2 = 1$. Finally, provisionally set r_3 to 1.

The stochastic enforcement of double cancellation works as follows. Let A be the Boolean variable representing the truth value of the statement $\hat{\theta}_{(2,1)} \leq \hat{\theta}_{(1,2)}$ and B the truth value of $\hat{\theta}_{(3,2)} \leq \hat{\theta}_{(2,3)}$. Then, if $i = 1$ and $j = 3$ there are two possible modifications:

- If the inequalities A and B from the above are both true, l_3 is set to $\hat{\theta}_{(3,1)}$.
- If the inequalities are both false, then r_3 is set to $\hat{\theta}_{(3,1)}$.

Note that in both cases the value that goes into l_3 or r_3 is the same, the question is whether it goes into the set of left- or right-hand limits. Similarly, if $i = 3$ and $j = 1$, then

- if A and B are both true, r_3 is set to $\hat{\theta}_{(1,3)}$ and
- if A and B are both false, l_3 is set to $\hat{\theta}_{(1,3)}$.

If the inequalities A and B have different truth values, then both r_3 and l_3 retain their provisional values.

Now define

$$f(l_1, l_2, l_3) = \max\{l_1, l_2, l_3\}.$$

The definition of g is more complicated. These complexities are due to the fact that there are few guarantees about how well behaved current values in the chain will be with respect to their ordering, but it is necessary for the jumping distribution to be always well defined (such that the left-hand value is strictly less than the right-hand value). Begin by defining

$$g'(r_1, r_2, r_3) = \begin{cases} \min\{r_1, r_2, r_3\} : r_3 > f(l_1, l_2, l_3), \\ \min\{r_1, r_2\} : r_3 \leq f(l_1, l_2, l_3). \end{cases} \tag{A.1}$$

Note that g' will always be defined as $\min\{r_1, r_2, r_3\}$ unless one is considering a corner cell. In that case, g' is defined in such a way to guarantee that a deviant r_3 value from the double cancellation check does not lead to a situation in which the left-hand side and the right-hand side of the uniform distribution "cross." This can be illustrated using the following hypothetical matrix:

$$\begin{matrix} a & b & c \\ d & e & f \\ g & h & i \end{matrix}. \tag{A.2}$$

Assume one is forming a jumping distribution for the cell with current value g. Assume that $d < b$ and $h < f$ (thus, double cancellation requires that $g < c$). The left-hand side of the uniform jumping distribution is going to be $f(l_1, l_2, l_3) = \max\{d, 0, 0\} = d$. Due to the assumptions just made regarding double cancellation, r_3 is set to c. Without the correction we would have $g'(r_1, r_2, r_3) = \min\{1, h, c\}$. The punchline is that since *no* ordering can be assumed along the minor diagonal, it might be the case that $c < d$. In that event, the jumping distribution could be malformed if $\min\{1, h, c\} = c$. The correction in Equation (A.1) is meant to account for such a situation.

Now, define

$$g(r_1, r_2, r_3) = \begin{cases} g'(r_1, r_2, r_3) : g'(r_1, r_2, r_3) \geq f(l_1, l_2, l_3), \\ 1 : g'(r_1, r_2, r_3) < f(l_1, l_2, l_3). \end{cases} \tag{A.3}$$

This modification is meant to handle additional potential problems with the corner cells. Again using the notation in Example (A.2), suppose we are forming the jumping distribution for the cell with current value c. Under appropriate assumptions about double cancellation, let $l_3 = g$ and the left-hand side of the jumping distribution is then $f(l_1, l_2, l_3) = \max\{0, b, g\}$. Since $r_3 = 1$ here, the correction of Equation (A.1) does not apply and we have $g'(r_1, r_2, r_3) = \min\{f, 1, 1\} = f$. However, it could well be the case that $g'(r_1, r_2, r_3) = f < g = \max\{0, b, g\} = f(l_1, l_2, l_3)$. Such an event would lead to a malformed jumping distribution without the correction of Equation (A.3). Although there is some asymmetry in that the right-hand limit is being treated slightly differently than the left-hand limit, this is not especially concerning since the right-hand limits are the ones that are still to be updated at each iteration of the chain.

Once the jumping distribution is defined, the Metropolis–Hastings algorithm is used as usual. A proposed value for the parameter is drawn from the jumping distribution. If $\hat{P}_{ij}^{t-1}$ is the last value in the MCMC chain and $\hat{P}_{ij}^{*}$ is the proposed value, then compute

$$r = \frac{(\hat{P}_{ij}^{*})^{n_{ij}}(1 - \hat{P}_{ij}^{*})^{N_{ij}-n_{ij}}}{(\hat{P}_{ij}^{t-1})^{n_{ij}}(1 - \hat{P}_{ij}^{t-1})^{N_{ij}-n_{ij}}},$$

where n_{ij} is the number of correct responses by the individuals in row i to item j and N_{ij} is the number of individuals in row i who responded to item j. Draw t from Unif[0, 1]. If $r > t$, then set $\hat{P}_{ij}^{t} = \hat{P}_{ij}^{*}$. Otherwise, set $\hat{P}_{ij}^{t} = \hat{P}_{ij}^{t-1}$. In each cell of the 3-matrix, 3000 iterations are performed of which the first 1000 are burned; and then the chain is thinned to only every fourth draw. The 95 % credible region for the parameter is then estimated using the remaining values. A distortion is detected here if

$$\hat{P}_{ij}^{\text{MLE}} = \frac{n_{ij}}{N_{ij}}$$

does not lie in this 95 % credible region. Note that nine Markov chains are being formed simultaneously within each 3-matrix.

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